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DanSto – Japanese Dataset of Short Stories for Evaluating Context Understanding

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Abstract: This paper introduces a dataset comprising more than 8,000 manually crafted short stories in the Japanese language. The primary objectives of this dataset encompass addressing the dearth of comparable data in Japanese. Additionally, the dataset provides alternative endings for the narratives through crowdsourcing, ensuring they remain both plausible and marginally less probable than the original ones. This approach contributes to the creation of a testing benchmark that poses heightened challenges for contemporary large language models, particularly when contrasted with analogous benchmarks in English where conclusions are typically dichotomized into correct and incorrect endings. The dataset is further expanded through automated manipulation of subjects and objects, and the study evaluates the performance of popular models across three key tasks: a) predicting story endings, b) substituting antonyms, and c) swapping nouns. Preliminary experiments show that zero-shot GPT-4 capabilities are relatively high, especially in case of recognizing sentences with swapped nouns (94% accuracy) while open-source Japanese LLMs struggle with processing proposed stories.

1 Introduction

Lack of commonsense understanding is one of the oldest problems of artificial intelligence, and despite the revolution in NLP related to the application of large language models (LLMs), it is rather hard to say that this problem has been solved. Although the common sense acquisition reaches beyond text, natural language processing is a leading field trying to equip machines with understanding the world in a deeper, more human-like manner. In recent years a plethora of corpora and datasets has been created to boost the advances in common sense processing, however most of these resources are available only for English language. Because of this problem, many NLP researchers are forced to work on English data which slows down the progress of implementing solutions in their own language.

In this paper we present a dataset which can be used for commonsense understanding in a wider span – five sentence stories. As there has been no available Japanese corpora of stories, we have created a dataset inspired by ROCStories [Mostafazadeh 16], a dataset of 100,000 short stories with more and less plausible endings called DanSto (Dangerous Stories). It contains 8,757 crowdsourced five sentence stories derived from DanSen (Dangerous Sentences) – 21,592 sentences related to safe and risky or hazardous acts [Katsumata 22], part of which became titles of the DanSto. Our decision to utilize this specific dataset as the base for stories lays in the fact that crowdsourced sentences are not published online to avoid data contamination (sentences are not available for training of large language models and other machine learning systems). DanSto differs from ROCStories not only in aspect of language and topic specificity, we also decided to add alternative story endings which are not clearly correct or wrong, but just slightly less plausible than the originals.

In the pursuit of Artificial General Intelligence (AGI), the ability to comprehend and interpret narratives holds paramount significance. Narratives, embedded within stories, serve as quintessential vessels of human experience, encapsulating cultural nuances, emotional intricacies, and complex contextual information. The comprehension of stories not only mirrors human cognition but also serves as a pivotal benchmark for the development of AGI systems. Understanding stories

involves extraction of underlying themes, the discernment of character motivations, the inference of implicit knowledge, and the synthesis of contextual information. Story understanding is required for engaging in meaningful interactions with humans and opens avenues for creative content generation. By providing examples how to alter the original dataset to test LLMs, we encourage researchers to utilize our dataset for further experimentation.

The paper is structured as follows. First we describe common sense-related datasets which have been previously made available in English and Japanese, and then we present details of our datasets together with simple baselines. We conclude the article with a discussion on possible uses of the data and share our plans for the future work.

2 Related Work

Narrative understanding is a challenging task for machines studied for decades [Winograd 72, Charniak 72]. As comprehending whole novels remains problematic to the field natural language processing, researchers developed shorter datasets of stories and knowledge bases which could be useful for the task of story understanding. Below we list examples of related works for English and Japanese languages.

2.1 Data in English

There are quite a few datasets that are commonly used for commonsense reasoning tasks in English. Traditionally they have been developed as knowledge graphs, and ConceptNet [Speer 17] is one of the oldest and most popular examples – although developed as multilingual, languages other than English are underrepresented[1]. It is open knowledge graph that represents general knowledge about the world, including common sense knowledge about people, objects, and events and has been used also in story understanding tasks [Williams 17, Huang 20]. WebChild [Tandon 17] is a dataset of web-based commonsense knowledge describing over 2 million disambiguated concepts and activities, connected by over 18 million assertions. Especially the activities knowledge can be directly used for story-related processing. ATOMIC [Sap 19a] is another large-scale dataset of human-annotated relationships between entities, such as "person sits on chair." It was developed by the Allen Institute for AI and COMET [Bosselut 19] is a ATOMIC-based model that can generate implicit commonsense inferences along several dimensions such as pre- and post-conditions, motivations, and mental states of the participants. To make a model that is more directly useful for story-related tasks, [Gabriel 21] have proposed PARA-COMET, a discourse-aware model that incorporates paragraph-level information to generate coherent commonsense inferences from narratives. Commonsense inference corpus for the Story Cloze Test (SCT) task has been proposed by [Yao 22]. The study gathered data from 1871 stories, with each story being labeled by three human workers. Intra-category and inter-category agreement analyses reveal substantial consensus. The authors also introduce two novel tasks: predicting narrative inference categories and identifying contributing sentences. Their findings demonstrate that transformer-based models achieve state-of-the-art (SOTA) performance on the original SCT task through transfer learning but struggle with more demanding tasks. CommonsenseQA [Talmor 22] is an example of a benchmark widely used to test LLMs. This is a dataset of multiple-choice questions that require the use of common sense to determine the correct answer. It was also developed by the Allen Institute for AI. SWAG [Zellers 18] is another large-scale benchmark of human-generated commonsense sentences that describe events and their effects developed by researchers at the University of Pennsylvania. The SWAG dataset consists of 113k multiple choice questions about grounded situations, including 73k for training, 20k for validation, and 20k for testing. The questions are video captions with four answer choices about what might happen next in the scene. The correct answer is the real video caption for the next event in the video, while the incorrect answers are adversarially generated and human verified to fool machines but not humans. The goal of the SWAG dataset is to play a role of a benchmark for evaluating grounded commonsense natural language inference and learning representations, but as most of such datasets it concentrates on short range of causal knowledge. This problem of simple pairs of sentences was one of the motivations for creating ROCStories [Mostafazadeh 16], a dataset of narrative-style commonsense stories that are designed to test a model's ability to understand and generate simple

[1] There are 12,397,055 assertions for English versus 1,631,492 for Japanese in ConceptNet 5.7.

stories based on common sense knowledge. The authors also provide alternate endings for some stories (Story Cloze Task) which inspired us while extending the DanSto dataset. StoryCommonsense [Rashkin 18] is a dataset derived from ROCstories and contains the annotations of motivations and emotional reactions of story characters. Sap and colleagues have proposed Social IQa [Sap 19b], a dataset of social scenarios that are designed to test if a system or model can understand and reason about social interactions and relationships based on common sense knowledge. It consists of 38,000 multiple choice questions aimed at testing emotional and social intelligence in everyday situations (e.g., Q: "Jordan wanted to tell Tracy a secret, so Jordan leaned towards Tracy. Why did Jordan do this?" A: "Make sure no one else could hear"). TruthfulQA [Lin 22] tested LLMs' capability to answer 817 questions in 38 categories, including health, law, finance and politics.

2.2 Data in Japanese

Compared to English resources, datasets for commonsense reasoning in Japanese are scarce. As mentioned before, although ConceptNet [Speer 17] contains assertions in Japanese, their number is approximately ten times smaller than these in English. There were attempts to extend the ConceptNet knowledge in Japanese but the acquired data contained too many factoids [Krawczyk 16], concentrated on only few relations [Shudo 16] or was not made public [Saito 18]. In 2019 Japanese version of GLUE [Wang 18], called JGLUE [Kurihara 22] has been released. It contains six benchmarks that can be indirectly helpful for story understanding, for example Japanese versions of popular sets like SQUAD [Rajpurkar 16] or CommonSenseQA [Talmor 22] are presented. [Omura 20] created a method for building a commonsense inference dataset that combines automatic extraction from a corpus and crowdsourcing. Each problem is a multiple-choice question that asks contingency between basic events. Recently, a commonsense knowledge graph on events in Japanese [村田 23] closely inspired by ATOMIC and COMET has been crafted from scratch, without translation, and released[2]. The seed graph has been crowdsourced and then expanded by in-context learning with a Japanese LLM. [土屋 22] developed a novel dataset of Japanese cloze questions tailored to assess machine reading comprehension. Derived from Japanese literature corpus, the dataset comprises questions automatically generated through a systematic process. Each question is represented as four parts – a contextual passage, a query featuring a placeholder, a designated answer character, and a pool of potential answer characters. The generation of queries follows a structured approach, originating from sentences immediately succeeding the context passages within the target literature. These queries are constructed by substituting the answer character with the placeholder, thereby fostering a comprehensive evaluation framework for machine reading comprehension in Japanese text. Although this dataset is the closest one to ours, it is based on existing novels and limits the use to the task of filling gaps. Note that similar approach can be also used in extending our corpus by removing subjects and adding alternative choices. Except the work of Katsumata et al. [Katsumata 22], mentioned before, there is a datasets in the realm of ethics named JCommonsenseMorality [竹下 23] which goal is to test LLMs' capability of recognizing immoral acts.

3 Dataset Creation

In this section we present how we created the DanSto dataset and its alternated subsets.

3.1 Main Story Dataset

As mentioned in the Introduction, we have utilized a corpus of sentences for automatic recognition of dangerous acts previously crafted by our group [Katsumata 22], which we call DanSen. Danger level of every sentence has been annotated by crowdworkers on the scale from 1 to 7. We randomly sampled 4,400 sentences which were annotated as "safe" by the majority of workers (5,321 sentences) and 4,400 which were evaluated as describing a dangerous act (4,436 sentences). Then we combined both sets and shuffled the merged dataset to expanded as a story corpus. We asked 44 crowdworkers[3] to create 220 five-sentence stories about assigned topics (sentences derived from DanSen as story titles). We chose annotators trying to keep balance between genders (24 females and 20 males) and ages (between 19 and 68 years old).

[2] `https://github.com/nlp-waseda/comet-atomic-ja`

[3] Creators were hired on `https://crowdworks.jp/` platform.

3.2 Alternative Ending Subset

The annotation guideline was simple and no restrictions about the content or length were made. The only condition was to at least use noun, particle and verb in a sentence. 9,000 JPY (before tax) has been paid for a set. The workers discovered various problems (e.g. incorrect ideogram use) in the titles, and after removing all problematic stories (43), the final DanSto set comprised 8,757 stories[4].

3.2 Alternative Ending Subset

To allow Story Cloze Test evaluation, we hired different 30 workers to create a little bit less plausible ending to a story. Each has been provided with 292 or 291 DanSto stories and asked to input a different last sentence which fits the context of the story, resembles the original ending and is not just its negation. These directives were made to avoid completely different endings which could be easily recognized by similarity measures. We avoided negation as it is a part of our separate study on evaluating negation understanding of LLMs. 6,600 JPY (before tax) has been paid for the addition to all 8,757 stories. We selected 15 females and 15 males, aged between 22 and 69 years old. Examples of the extended DanSto dataset are shown in Table 1.

In the next step we hired other 5 workers to confirm whether the proposed alternative endings were really slightly less probable than originals. We chose 464 stories which both endings (original and added) have the same number of characters to minimize influence of the string length while calculating similarity or perplexity. These stories are later used as a test set in our experiments. We paid 3,800 JPY (before tax) and the workers (two females and three males) aged between 33 and 55 years old. The agreement percentage between the labelers is presented in Table 2.

3.3 Swapped Antonyms Subset

In order to extend the range of experiments for LLMs, we decided to alternate the DanSto dataset further. The first subset has been created by utilizing an antonym list[5] to change a single word in a story sentence if it has an antonym. This is to examine if a model can recognize that one of the sentences within a story (titles remained unchanged) can be discovered as unnatural one causing any illogicality. We limited the swapping process to one noun and if there were two or more candidates for replacement or there were two or more possible antonyms, the choice was random. In the end one random sentence with the replaced word was chosen to be the unnatural one to be discovered. We processed tokens, not single character antonyms as they sometimes changed existing words to non-existing ones. For example "美味しい" (*tasty*) would be replaced by "醜味しい" (where ideogram for *beautiful* taste is changed to *ugly* taste). As humans could comprehend the meaning of a word created this way, testing LLMs recognition of this kind of new words might be a topic for future research. In Table 3 we present an example of transformed story with a replaced antonym.

3.4 Swapped Nouns Subset

Another alternative subset was created by reordering subjects and objects positionally. To achieve this we parsed sentences with SpaCy [Honnibal 20] and by using similar strategy as with antonym swapping, we randomized choices if more than two candidates existed. An example of the resulting object-subject reordering is given in Table 4.

4 Experiments and Results

For the experiments we use two recent models of Generative Pre-trained Transformer (GPT), namely *gpt-4-0613* and *gpt-4-0125-preview*[6], as they were among top five LLMs as ranked by human preference in the time when the test were conducted[7]. For all the experiments we used temperature=0, empty system message and maximum output tokens set to 1. All prompts were in English and they are reported together with setup details in the Appendix.

[4]The dataset is not currently available online to avoid data contamination but can be provided upon request by e-mailing the first author.

[5]https://github.com/sowa-s/JapaneseAntonymSearcher/blob/master/antonym.csv

[6]https://platform.openai.com/docs/models/overview

[7]https://huggingface.co/spaces/lmsys/chatbot-arena-leaderboard, accessed on February 14, 2024.

4.1 More Plausible Ending Identification

Table 1: Story examples (original and translated to English)

Title	Story	Ending 1	Ending 2
選手が敵の作戦が書かれたノートを刃物で脅して奪う	選手は敵に勝ちたかった。選手は敵の作戦ノートがあることを知った。選手は奪い取ろうと思った。選手はそれを強奪した。	選手は強盗罪で逮捕された	選手はそれでも勝てなかった
A player threatens and steals a notebook containing the enemy's strategy with a knife.	The player wanted to win. The player learned about the enemy's strategy notebook. The player decided to snatch it. The player succeeded in robbing it.	The player was arrested for robbery.	Still, the player couldn't win.
人が事務所にあるテープカッターを使用してテープを切る	事務所で作業している人がポスターの張替えを頼まれる。貼ってあるポスターを剥がす。丸まっているポスターを伸ばす。事務所にあるテープカッターを使いテープを 4 回切る。	ポスターの四隅にテープでとめポスターを貼る。	その人はうっかり指を深く切ってしまった。
A person uses a tape cutter in the office to cut tape.	A person working in the office is asked to replace a poster. They peel off the existing poster, stretch the rolled-up poster, and use the office tape cutter to cut the tape four times.	They tape the poster to the four corners.	The person accidentally cut their finger deeply.

Table 2: Raters agreement ratio for 464 test set stories

Agreement Level	**Count**	**Percentage**
5 annotators agreed	266	57.32%
4 annotators agreed	117	25.23%
3 annotators agreed	81	17.45%

4.1 More Plausible Ending Identification

We prompted both OpenAI models to recognize more probable ending for all 464 stories from the test set. As shown in Table 5, the older model, namely *gpt-4-0613* has recognized correctly significantly more plausible endings than its more recent equivalent, *gpt-4-0125-preview*[8].

4.2 Illogical Sentence Identification

We used the same prompt using 100 random stories with antonym-replaced sentences, and 100 random stories for subject and object-replaced sentences. We have examined both *gpt-4-0613* and *gpt-4-0125-preview* models. As models sometimes refused to choose any of the sentences (judging all of them as logical), we extended the prompt (see Appendix). The results are presented in Table 6.

[8]During testing, the order of A and B was randomized.

4.3 Experiments with Japanese LLMs

To compare OpenAI models with existing open-source large language models dedicated to Japanese, we have tested the same zero-shot approach for the main story cloze task using following models (accuracy given in brackets), basically reusing the code available on their HuggingFace pages:

- japanese-stablelm-instruct-alpha-7b (56.46)[9]
- llm-jp-13b-instruct-full-jaster-v1.0 (48.70)[10]
- Swallow-7b-instruct-hf (64.87)[11]
- weblab-10b-instruction-sft (57.11)[12]

As models often did not generated properly formatted output (or most of outputs were identical as in the case of LLM-jp model), we experimented with various English and Japanese prompts, however only performance of the Swallow-7b model visibly exceeded random choice when a mixed English-Japanese prompt was used (see Appendix). Therefore we used it for experiments on two additional subsets, from which 100 stories each were randomly selected. Howeve,r the model was always selecting the first sentence (output=1) in spite of testing English, Japanese and bilingual prompts. Further experiments with other parameter configurations and models are needed, also

[9]`https://huggingface.co/stabilityai/japanese-stablelm-instruct-alpha-7b` accessed on February 17, 2024

[10]`https://huggingface.co/llm-jp/llm-jp-13b-instruct-full-jaster-v1.0` accessed on February 17, 2024

[11]`https://huggingface.co/tokyotech-llm/Swallow-7b-instruct-hf` accessed on February 17, 2024

[12]`https://huggingface.co/matsuo-lab/weblab-10b-instruction-sft` accessed on February 14, 2024

Table 3: Example of a story with a noun replaced by antonym

Original	**Translation**
「相手が料理をするためにナイフを使う」	"The partner uses a knife to cook"
私は明日出張で家を空ける予定だ。	I am planning to leave home on a business trip tomorrow.
食事作りのために夫に調理を教えることにした。	I decided to teach my husband how to cook for meals.
夫はたどたどしくナイフを使い食材を切った。	Husband awkwardly used the knife and cut the ingredients.
食材はいびつな形に切られた。	The ingredients were cut into irregular shapes.
妻は心の中で（自分→**他人**）を誇らしく思った。	The wife felt proud of (herself → **somebody else**) in her heart.

Table 4: Example of a story with a subject replaced by an object

Original	**Translation**
「相手が料理をするためにナイフを使う」	"The partner uses a knife to cook"
私は明日出張で家を空ける予定だ。	I am planning to leave home on a business trip tomorrow.
食事作りのために夫に調理を教えることにした。	I decided to teach my husband how to cook for meals.
ナイフはたどたどしく**夫**を使い食材を切った。	**Knife** awkwardly used the **husband** and cut the ingredients.
食材はいびつな形に切られた。	The ingredients were cut into irregular shapes.
妻は心の中で自分を誇らしく思った。	The wife felt proud of herself in her heart.

Model	Ending A	Ending B	Accuracy
gpt-4-0613	373	91	**0.803**
gpt-4-0125-p	345	119	0.743

Table 5: Accuracy of OpenAI GPT models on more probable ending identification (Ending A is the one chosen by majority if human voters)

Subset	Model	Count	Accuracy
Antonyms	gpt-4-0613	76/100	76%
	gpt-4-0125-p	78/100	**78%**
Subj-Obj	gpt-4-0613	93/100	93%
	gpt-4-0125-p	94/100	**94%**

Table 6: Accuracy of OpenAI GPT models on illogical sentence identification

testing few-shot approach or fine-tuning should improve the performance of smaller, more open and free models.

5 Analysis and Discussion

As mentioned above, results of the main experiment showed that *gpt-4-0613* yielded 6 points higher accuracy than *gpt-4-0125-preview* model, suggesting that for choosing more plausible story endings, older model provided by OpenAI can be more fit to this kind of tasks, at least when they are written in Japanese. 80% of correct predictions might be considered as moderate knowing how well GPT-4 models can perform on various in-context learning tasks, however, it should be noted that the specificity of the additional story ending (being only little less probable than original), makes the task challenging. As Table 2 shows, less than 60% of evaluators agreed completely which ending of a story is less plausible. For that reason we performed further analysis of how this difficulty differs between humans and models and found a distinct correlation. GPT-4 accuracy for endings agreed by three annotators was 45.68%, for four 72.65% and 83.83% for cases where all five human evaluators agreed. Further analysis might also reveal some average personality or experience-oriented tendencies inherited by models from different data. For example, in the following story titled "Man softly hits a pillow", GPTs and Swallow chose different endings as more probable:

> I had nothing to do, I was bored. Somehow, a pillow caught my eye. Somehow I found myself touching the pillow. As I touched the pillow, I thought about what I can do with it.

The ending the GPT-4 selected as more probable was "I couldn't think of anything, so I lightly hit the pillow", while Swallow regarded "I couldn't think of anything, so I put the pillow back." as more plausible ending. All Japanese evaluators agreed that the first ending is more probable, while model trained heavily on Japanese texts did not agree. Such dependencies could be studied further, however many more evaluators are necessary for drawing any meaningful conclusions.

Regarding perturbed sentences in the stories, recog-

nition performance varied depending on the subset. Results showed clearly that reordering words is much easier to be noticed by the GPT-4 models than replacing them with antonyms. Nevertheless, it has to be noted that human evaluation of replacements in sentences will be necessary as they were done automatically and chosen randomly without human intervention. Some word-replaced sentences were still meaningful after the alternation, but the whole story was to disambiguate such situations. In the following story "In the theater daughter plays Snow White and eats poison apple", the last sentence is such an example:

> I went to my daughter's kindergarten presentation. It was "Snow White" and my daughter was the main character. My daughter ate a poisoned apple and fell asleep. I got bored watching the play because she didn't play anymore. To kill smartphone I was looking at time.

The last sentence (時間を見てスマホをつぶした) is quite natural in Japanese as it means to (incidentally) crushed a smartphone while looking at the time, but the previous sentences explain why it should be "To kill time I was looking at smartphone". The results suggest that this task is easier to solve for GPT-4 than identifying sentences with antonyms.

As we mentioned earlier, further experiments with non-commercial Japanese LLMs are needed to fairly examine their ability for recognizing perturbed sentences as the authors did not manage to obtain meaningful results in the zero-shot scenario.

6 Conclusions and Future Work

In this paper we proposed a novel, original a corpus of original short stories for Japanese language. Unrestricted event understanding in any context is a capability that any AGI must posses, therefore stories we introduced in this paper are relatively free when it comes to length or building up the plot from a title. Sentences forming a story are short as there was no incentive for crowdworkers to be elaborative. This means that the five sentences are a very basic set of event descriptions allowing human readers to understand what happened in the story. One of our intentions for creating this dataset was to allow studies on how current and future algorithms will interpret such compressed chains of events or how these events can be decompressed. How they can be depicted by images or videos, how changing small elements could cause unexpected new negative and positive storylines. For time being we have only conducted basic tests with LLMs to provide baselines to be further improved. In the near future we are going to concentrate on comparing how different is humans and machines ability to explain by unpacking the tacit knowledge hidden inside the sentences to discover possible dangers and plan ethical countermeasures. We have also started using this dataset for automatic story ending generation and its discovery, the corpus is also being utilized for humorous ending generation as a part of creativity-related research.

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Appendix

Prompt for More Plausible Ending Identification (GPT-4)

Here is a story in Japanese titled {title}:
{sentence1}
{sentence2}
{sentence3}
{sentence4}
Which ending is more probable?
A) {ending1}
or
B) {ending2}
Answer only with A or B.

Prompt for Illogical Sentence Identification (GPT-4 / Swallow)

Here is a story in Japanese titled {title} consisting of the following five sentences:
{sentence1}
{sentence2}
{sentence3}
{sentence4}
{sentence5}

Identify the most illogical sentence in the context of the entire story and answer only with a single number corresponding to the number of that sentence. Never answer with anything else, like for example by saying that there is no illogical sentence in the story. There is precisely one illogical sentence there. Even if you do not know which sentence is that for certain, you must answer only with a single number corresponding to the sentence most likely to be illogical.

Prompt for More Plausible Ending Identification (Swallow)

Below is a story titled title. Read the story and choose more plausible ending. You can answer only by choosing 結末 A or 結末 B.
story
結末 A：ending1
結末 B：ending2
Most plausible is 結末